

Profile

I am a backend engineer with a solid foundation in **Python, Flask, and Docker**, along with practical experience in **TypeScript and C**.

I prioritize engineering practices focused on **scalability, reliability, and maintainability**, thriving in environments where technology and **collaboration drive lasting impact**.

In addition to development, I have hands-on experience building CI/CD pipelines using both GitLab and GitHub, ensuring reliable delivery workflows and consistent deployment processes. While I am comfortable working across the stack, my core strengths lie in **backend engineering, framework design, and service architecture**.

Currently, I am expanding my expertise in **Azure Cloud** and Configuration Management with **Ansible**, bridging the gap between **development and operations**. This growth reflects my passion for creating not only excellent applications but also the **resilient, scalable infrastructure** that supports them.

Work History

Senior Software Engineer
Heidelberg Materials Digital Hub Varna
04/2025 – present Varna, Bulgaria

Tech & Tools

- PxTrend, Python, Azure, Docker, Ansible, Jenkins, Git, GitLab, Elasticsearch, Grafana, Kibana, JIRA, Confluence

Core Skills

- OOP, SOLID Principles, Design Patterns, Agile Principles, Customer-Centric Delivery, Cross-Functional Collaboration, Technical Documentation

Key Achievements

- Provided ongoing support for PxTrend (data historian), fixing bugs and refactoring components to meet evolving business needs
- Contributed to root cause analysis and postmortem documentation following operational incidents
- Refactored ETL pipelines for reliable data transfer from OT to IT systems
- Proactively engaged plant operators and end users on-site to identify and resolve real operational pain points

Work History

- Handled customer issues and requests, delivering timely resolutions and maintaining service continuity
- Authored technical documentation in arc42 format to improve maintainability and knowledge transfer
- Developed Ansible playbooks to automate monitoring and provisioning processes
- Created GitLab CI/CD pipelines to streamline service integration and improve release reliability

Senior Python Developer
Hilscher Development and Test Center
2019 – 04/2025 Varna, Bulgaria

Tech & Tools

- netX, Python, Docker, Git, SVN, JIRA, Confluence, Bitbucket, GitHub

Core Skills

- OOP, SOLID principles, Design Patterns, Software Architecture, Lean Agile Development, Technical Documentation, Mentoring

Key Achievements

- Authored production-quality industrial protocol libraries in Python (MQTT, DeviceNet, OpenModbus over TLS, etc.)
- Authored comprehensive and scalable test frameworks ensuring high quality products before system tests
- Refactored legacy code into a maintainable, scalable architecture by applying clean code principles
- Created and maintained training programs for new recruits, covering topics such as Python, OS, multi-threading, version control, and testing guidelines.
- Authored and maintained executive-level product documentation, ensuring clarity, alignment, and informed decision-making

Industrial Automation Engineer
Solvay Sodi
2009 – 2019 Devnya, Bulgaria

Tech & Tools

- Siemens, Honeywell, Profibus, Profinet, Open Modbus, Matlab, SciLab, MiniTab, Java, Visual Basic, C#, Siemens SCL

Core Skills

- Process Automation with Siemens, ATEX/IECEx Compliance, Human-Machine Interfaces. Process Modeling and Optimization, PID setup and tuning,



Work History

Advanced Process Control (APC)

Key Achievements

- Adapted Honeywell control blocks for use with Siemens S7 PLCs using Structured Control Language (SCL)
- Created real-time cooling towers simulation based on 1D Merkel models with own implementation of psychometric libraries in Visual Basic
- Implemented a Mamdani Fuzzy Logic Controller to optimize the production of carbonisation columns
- Automated a fly-ash installation in a boiler plant including HMI
- Automated a brine purification sector with Siemens PLCs including custom libraries in SCL and HMI
- Implemented tools to check and link P&ID diagrams using Visual Basic and Excel
- Implemented a custom logic to mitigate quality issues caused by flow transmitters that did not compensate for temperature above 37 degC

Software Developer Intern

X2E GmbH

2005 – 2007 Karlsruhe, Germany

Tech & Tools

- C, VHDL, Ruby

Core Skills

- OOP, Design Patterns, Automated Testing

Key Achievements

- Implemented MOST-protocol demonstration platform using VHDL and C
- Implemented a flash library for firmware updates using C and C++
- Implemented an automatic testing system for company products using Ruby



Education

Systems Engineering | Diplom-Ing. (TH)

Karlsruhe Institute of Technology (KIT)

2002 – 2008 Karlsruhe, Germany



Publications

Design and industrial implementation of fuzzy logic control of level in soda production

Engineering Science and Technology, an International Journal

<https://shorturl.at/pZtnc>

Snejana Yordanova, Milen Slavov, Branimir Georgiev. Vol 23, Issue 3, pages 691-699



Publications

Parallel distributed compensation for improvement of level control in carbonization column for soda production

Control Engineering Practice

<https://linkinghub.elsevier.com/retrieve/pii/S0967066117302277>

Snejana Yordanova, Milen Slavov, Branimir Georgiev. Vol 71, pages 53-60

Optimization of Parallel Distributed Compensation for Real Time Control of Level in Carbonization Column

Automatic Control and Computer Sciences

<https://link.springer.com/article/10.3103/S0146411620050107>

Snejana Yordanova, Milen Slavov, Branimir Georgiev. Vol 54, pages 379-390



Certifications

Microsoft Certified: Azure Fundamentals

Microsoft

01/2026

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Issued Jan 2026

Certified SAgile 6 Practitioner

Scaled Agile, Inc.

Issued Feb 2024 - Expires Feb 2025

Experion PKS C300 Control Execution and Environment

Training Center Prague

11/2010

S7PRO1 - Simatic S7 TIA Programming 1

Siemens Authorized SIMATIC Training Center Bulgaria

06/2010

1726/2010

S7PRO2 - Simatic S7 TIA Programmin 2

Siemens Authorized SIMATIC Training Center Bulgaria

02/2011

1815/2011



Technical Skills

Software Engineering	Expert
Quality Engineering	Expert
Python Development	Expert
Docker Containers	Advanced
CI/CD Pipelines	Skilled
C, Go	Skilled



Professional Skills

Project Ownership	Expert
Documentation	Expert
Mentoring	Advanced
Training	Advanced



Languages

Bulgarian	Native
English	Professional
German	Limited
French	Limited